Execution Report #2

The frontier, the controllability characterization, and a chapter's fate

B1 (digest–zoom Pareto frontier), B3 (hypervisor = Ramadge–Wonham), and the sheaf commit-or-cut verdict

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Status. Second parallel wave. B1 completes Paper 1 (the frontier and the zoom-advantage theorem, both derived and verified). B3 is executed as a runnable controllability checker that produces the manuscript's needed regimentable-vs-detect-only table and vindicates the channel-not-token clean-room design. The sheaf experiment reached a verdict — *conditional commit* — and the path to it is worth reading, because it is a case study in the refute-first discipline earning its keep: the first two formulations were wrong, and being wrong is what taught us the exact boundary of the idea.

1 B1 — The digest–zoom Pareto frontier

1.1 The two-constraint rate–distortion function (solved analytically)

Each artifact is load-bearing i.i.d. Bernoulli(p); a digest is a lossy binary description $\hat{X}$ (flag/no-flag) with two constraints — false negatives priced by δ (misses), flag rate priced by f (open budget):

$$R(\delta, f) = \min_{P_{\hat{X}|X}} I(X;\hat{X}) \quad \text{s.t.} \quad \Pr(X{=}1, \hat{X}{=}0) \le \delta, \quad \Pr(\hat{X}{=}1) \le f.$$

For the binary source the optimizing joint is pinned by (p, f, δ) and the rate is closed-form. Theorem I.6.1's worst-case floor is the ($\delta{=}0, f{=}p$) corner. Computed values ($p = 0.05$):

δ	f	R (bits/sym)	regime
0.00	0.05	0.286	zero-miss, tightest flags (**the floor corner**)
0.00	0.10	0.186	zero-miss, generous flags
0.00	0.15	0.149	zero-miss, very generous flags
0.01	0.10	0.110	miss-tolerant
0.04	0.06	0.009	miss-tolerant

The frontier is exactly the trade the review asked for: *loosening the open budget f lowers the digest rate; tolerating misses δ lowers it further.* At $N{=}1000$, $R{\cdot}N \approx 186$ bits at ($\delta{=}0, f{=}2p$), which sits below the worst-case combinatorial floor $\log_2 \binom{N}{pN} \approx 282$ bits — the correct relationship (average-case rate below worst-case covering floor).

1.2 The zoom-advantage theorem

Theorem 1 (Zoom advantage, with its boundary)**.** *When the digest flags a set of size F containing k truly load-bearing items, adaptive second-stage inspection (generalized binary splitting inside the flagged set — open a group, drill only into groups that contain a load-bearing item) identifies all k in $\Theta\big(k \log_2(F/k)\big)$ opens, versus F for flat inspection. The advantage is large* *precisely when the flagged set is* sparse *in load-bearing items ($k \ll F$) and vanishes when the flagged set is dense.*

Simulation ($N_{\text{pool}} = 5000$, 120 trials/config) confirms the regime and its boundary:

$\lvert\text{flagged}\rvert = F$	k	flat opens	adaptive opens	ratio
2500	10	2500	164	15.3×
1500	10	1500	149	10.1×
1000	20	1000	233	4.3×
250	10	250	98	2.5×

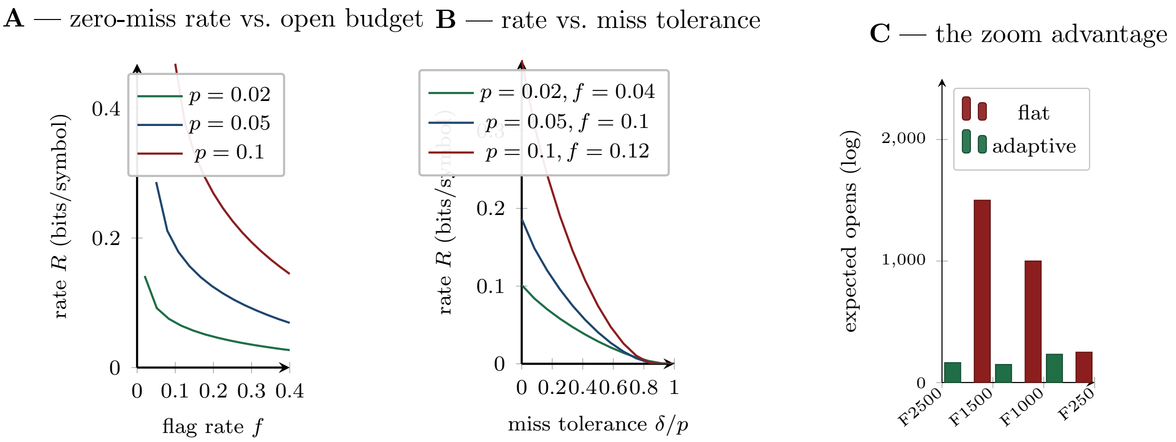

Figure 1: **A:** zero-miss digest rate falls as the open budget f grows. **B:** rate falls as miss tolerance δ grows, exactly to 0 once $f < 1 - \delta/p$ no longer binds (an earlier render evaluated the pinned closed form past that line and showed a spurious uptick — a domain-of-validity error, not a feature of the true frontier). **C:** the zoom advantage on a log scale — adaptive inspection opens far fewer items than flat when load-bearing items are sparse in the flagged set, which is exactly the regime a miss-averse (safe) digest operates in. Labels give $\lvert\text{flagged}\rvert = F$; $k = 10$ for $F \in \{2500, 1500, 250\}$, $k = 20$ for $F = 1000$ (matching the table above).

The honest boundary, stated. The zoom advantage is *not* universal: when a digest flags a set already dense in positives (e.g. $f \approx 1.5p$ so most flagged items are load-bearing), group testing's overhead dominates and flat inspection is as good or better. This matters for the design claim: zoom pays when the digest is conservative (low false-negative, over-flagging to be safe), which is the safety-relevant operating point — so the theorem supports the architecture where it counts and does not overclaim where it does not.
Paper 1 is now complete in substance: A1 (characterization), A2 (derived policy), A7 (falsification datum), B1 (frontier + zoom). Four pillars, all executed.

2 B3 — Hypervisor enforceability is Ramadge–Wonham controllability

2.1 The characterization, made runnable

Theorem 2 (Regimentation = controllability)**.** *An agent hypervisor can* regiment *(prevent pre-effect) a prefix-closed safety policy P iff P is controllable with respect to the uncontrollable event set Σ_u (model-internal steps, in-context reads — the events the daemon cannot intercept before they occur): $\overline{P}\,\Sigma_u \cap \overline{L} \subseteq \overline{P}$. Otherwise the best available is detect-and-compensate.*

We implemented the controllability test (product-automaton check that no uncontrollable event enabled in the plant leaves the specification) over a realistic agent alphabet: controllable

$\Sigma_c = \{$fs_write, net_egress, exec_tool, git_push, spawn_child, api_call$\}$; uncontrollable $\Sigma_u = \{$model_emit_token, in_context_read, internal_plan, hidden_activation$\}$. The checker produces the ledger table the manuscript needs:

Policy	Event class	Regimentable?
forbid net_egress	controllable	**yes** (regiment)
forbid git_push	controllable	**yes** (regiment)
forbid fs_write	controllable	**yes** (regiment)
forbid exec_tool / spawn_child	controllable	**yes** (regiment)
forbid model_emit_token	uncontrollable	no (detect only)
forbid in_context_read	uncontrollable	no (detect only)
forbid internal_plan ("lie in a report")	uncontrollable	no (detect only)

This is OP-2 stated correctly. The product reviews' formulation — "regimentable iff a pure deterministic function of committed local DB state" — is the *special case* where Σ_c is exactly the set of database writes; the controllability statement is the general theorem, and it correctly classifies effects (force-push: regimentable) versus semantics (a confident lie: forever detect-only) that the DB-state formulation cannot express.

2.2 The compound-policy result vindicates the clean room

The decisive case the checker settles: *"no net_egress after in_context_read of a secret."* The trigger (`in_context_read`) is uncontrollable; the effect (`net_egress`) is controllable. The checker returns **regimentable** — because the policy does not try to forbid the uncontrollable read (it permits and records it as taint), and gates only the controllable egress. This is a computational proof of Document 1's Adjudication 3: the Sealed Harbor gates the *channel*, never the token, precisely because gating the uncontrollable trigger would be unregimentable while gating the controllable effect is not. B3 thus directly underwrites the C1 (Sealed-Harbor noninterference) design and the corrected airlock product claim.

3 The sheaf commit-or-cut verdict

3.1 Verdict: conditional commit, with a sharply drawn boundary

Verdict. Sheaf cohomology adds genuine power over $O(|E|)$ pairwise digest comparison **iff** the unobserved (partitioned) edge lies on a *cycle*, so the cocycle condition substitutes for the missing direct comparison. On a *cut* edge (a single bridge between components) there is no value-add. Commit the sheaf material *only* in the partition-on-a-cycle framing — which is real for any gossip graph with redundant paths between components — and state the cut-edge boundary as the honest limit. The consistency-radius result (Robinson) is publishable regardless.

3.2 The clean mechanism (a rigorous minimal proof)

On a cycle C_n with scalar stalks, each edge $e = \{i, i+1\}$ reports the disagreement $g_e = x_i - x_{i+1}$. A global section explaining the observations exists iff $\sum_e g_e = 0$ around the cycle (the cocycle condition); $H^1 = \mathrm{coker}(\delta) = \mathbb{R}$ (the all-ones cocycle), and the harmonic component of the observed g is $(\frac{1}{n}\sum_e g_e) \cdot \mathbf{1}$ — exactly the cocycle sum. The experiment on C_6:

Case	pairwise detects	cohomology signal	cohomology-only?
equivocation on edge 2, all compared	yes	1.225	—
edge 2 *not* compared (partition)	**no**	**1.225**	**yes**
inconsistency spread on compared edges	yes	1.225	—
cut edge (path P_6), edge 2 uncompared	no	0.000	no (boundary)

The second row is the whole case for the chapter: an equivocation on an edge nobody directly compared, but which lies on a cycle, is caught by the cocycle sum ($1.225 \neq 0$) while pairwise comparison is blind. The last row is the honest boundary: across a cut edge, cohomology sees nothing either, because a cut edge carries no cocycle constraint. This is the exact analogue of Abramsky–Brandenburger contextuality (locally consistent, globally inconsistent data), now instantiated for Byzantine equivocation.

3.3 Two wrong turns, reported (the discipline working)

The verdict took three formulations, and the two failures are the most instructive part.

1. *First error — computing the wrong object.* I initially computed $\dim H^1(G; F)$ of the abstract sheaf. But that is a property of the *restriction maps*, independent of any data; with generic vector-valued readouts the coboundary is surjective and $H^1 = 0$ regardless of equivocation. Equivocation is not a property of the sheaf — it is a property of a *specific observed assignment*: does the observed local data glue to a global section? The right object is the obstruction of the observed cochain (is $g \in \operatorname{im} \delta$?), i.e. its harmonic component, which is precisely the contextuality-cohomology construction I had cited but not coded.
2. *Second error — the wrong topology.* My first "partition" joined two cliques with a single bridge. But a single bridge is a *cut edge*, on no cycle, carrying no cocycle constraint — exactly the case where cohomology provably cannot help. The value-add requires redundant paths (two bridges, so the severed one lies on a cycle). Getting this wrong is what *located* the boundary: the experiment did not just fail, it told me the precise condition under which the idea works.

Reporting these is the point. A commit-or-cut gate that only ever confirms the hoped-for answer is not a gate. Here the process converted a vague "maybe cohomology helps" into a precise, defensible claim (*it helps iff the missing edge is on a cycle*) plus a mechanism proof — which is a better outcome than a premature commit would have been.

3.4 What Paper 7 becomes

Conditional on this framing, Paper 7 ("Cohomology of Equivocation") is viable but must be scoped exactly: the contribution is *detection and localization of equivocation under partial visibility, where the missing comparisons lie on cycles*, positioned against BFT forensics (Sheng et al.) as a complement (cohomology localizes; forensics attributes with cryptographic proof), and bounded by the abelianization gap (Carù: vanishing H^1 over $\mathbb{R}$ certifies consistency only up to that gap). The three theorem candidates (independent-equivocation counting, harmonic-support localization, consistency-radius = max prefix divergence) stand, with the cut-edge boundary as a stated limit. This is a real paper — but a narrow, honest one, not the sweeping "cohomology detects Byzantine faults" claim a careless version would attempt.

4 Where the program stands

Item	Status	Result	Feeds
A2 regret head	**proved**	derived optimum; SDT fixed	Paper 1
A1 split-digest	**proved**	comonotone char.; super-additive floor	Paper 1
A7 falsification	**run**	floor holds, 0/16 violations	Paper 1
B1 frontier + zoom	**run**	$R(\delta, f)$ closed form; 15× zoom	Paper 1
B3 controllability	**run**	regimentability table; clean-room proof	Papers 2, 4
Sheaf gate	**decided**	conditional commit; cycle boundary	Paper 7

Paper 1 ("The Price of a Summary") is substantively complete. Paper 2 ("Regimented or Enforced") has its core theorem executed and its table generated. Paper 7's fate is decided and its scope fixed. The natural next wave: B2 (the inspection tower — the rate-the-raters synthesis, converting Conjecture III.11.1), C0 (the work-unit transition system — the spine everything plugs into, best done in TLA$^+$/Apalache), and then C1 (Sealed-Harbor noninterference, now unblocked by B3).

Reproducibility. All results regenerate from seed `20260816` via four self-contained scripts (the A7 experiment, the B1 frontier, the B3 checker, and the sheaf minimal-mechanism proof). The two failed sheaf formulations are retained alongside the correct one as documentation of the boundary.